Experiment :-7

Problem Statement:-

Implement a sliding window protocol with piggybacking for efficient data transmission and error control. Simulate data transfer between two nodes and visualize the window movements and acknowledgments using Cisco Packet Tracer.

Aim:- **Implementation of Sliding Window Protocol with Piggybacking**

Procedure:-

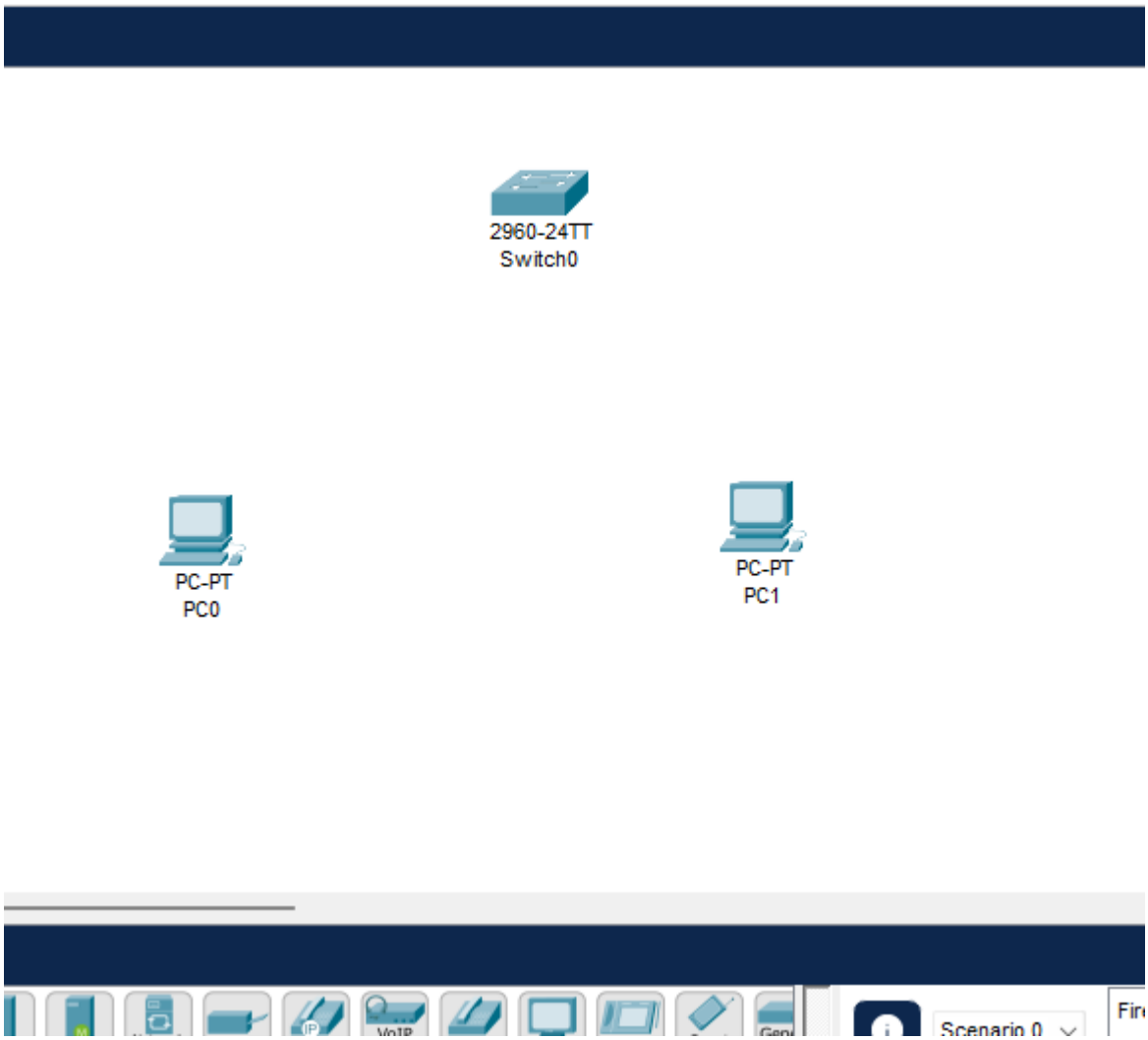
ART 1: Setup the Network Environment

Step 1: Open Cisco Packet Tracer

1. Launch Cisco Packet Tracer
2. You'll see a blank workspace in the middle

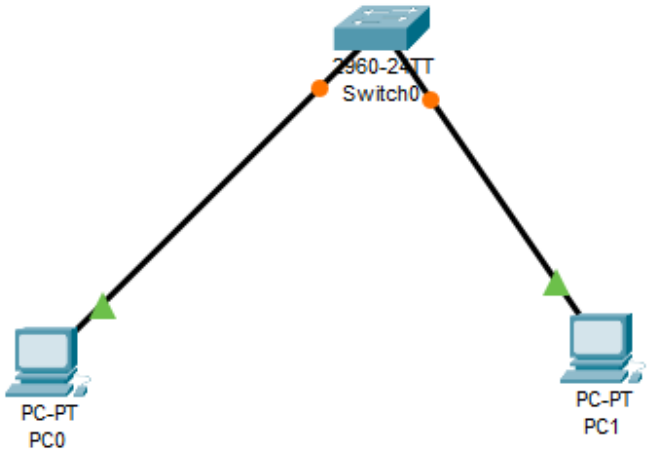
Step 2: Add Two PCs and one switch

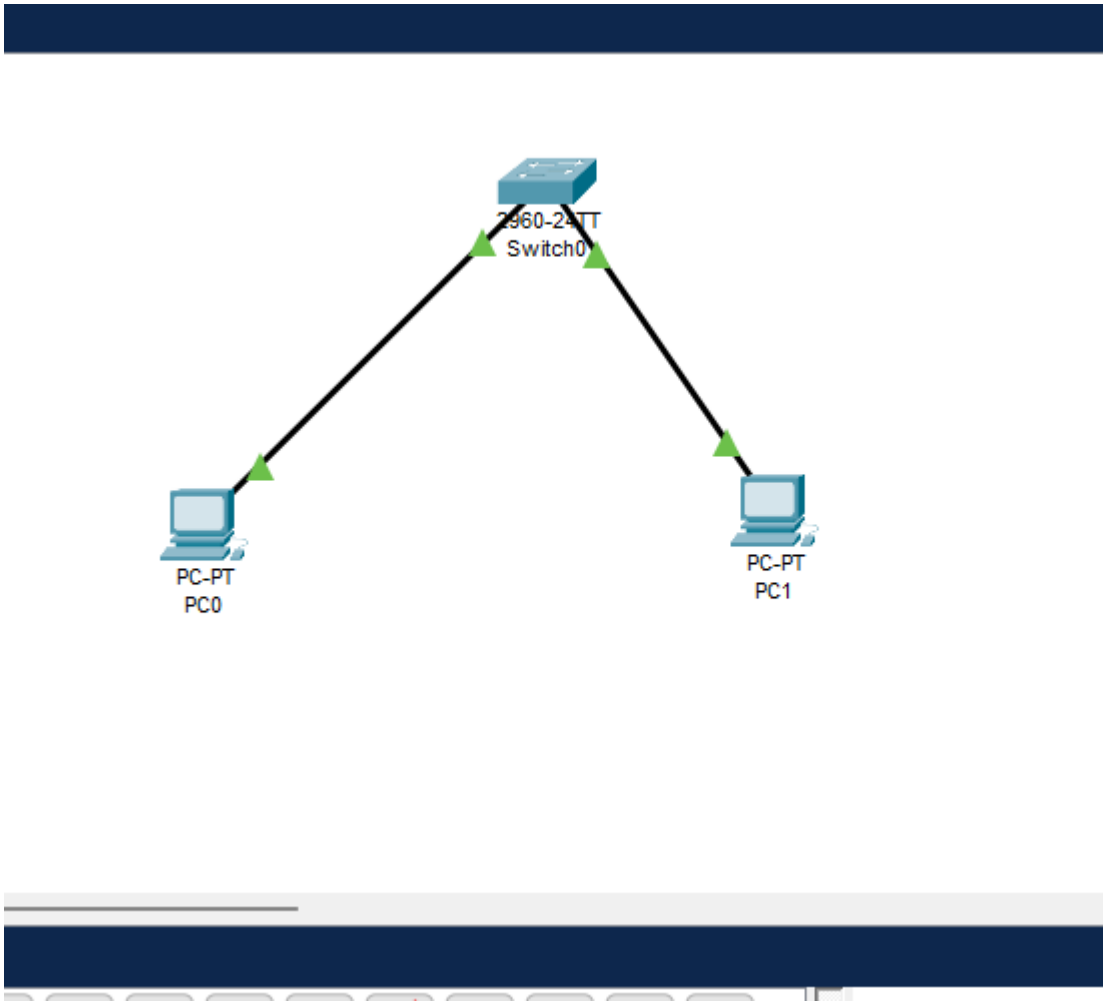
1. At the bottom panel, click **"End Devices"** (PC icon)
2. Drag and drop **PC0** onto the workspace (left side)
3. Drag and drop **PC1** onto the workspace (right side)
4. Leave some space between them



Step 3: Connect the PCs

1. Click **"Connections"** at the bottom (lightning bolt icon)
2. Select **"Copper Cross-Over Cable"** (the dashed line)
3. Click on **PC0** → select **FastEthernet0**
4. Click on **PC1** → select **FastEthernet0**
5. You should see a cable connecting both PCs
6. Wait for the link lights to turn **Green** ✅

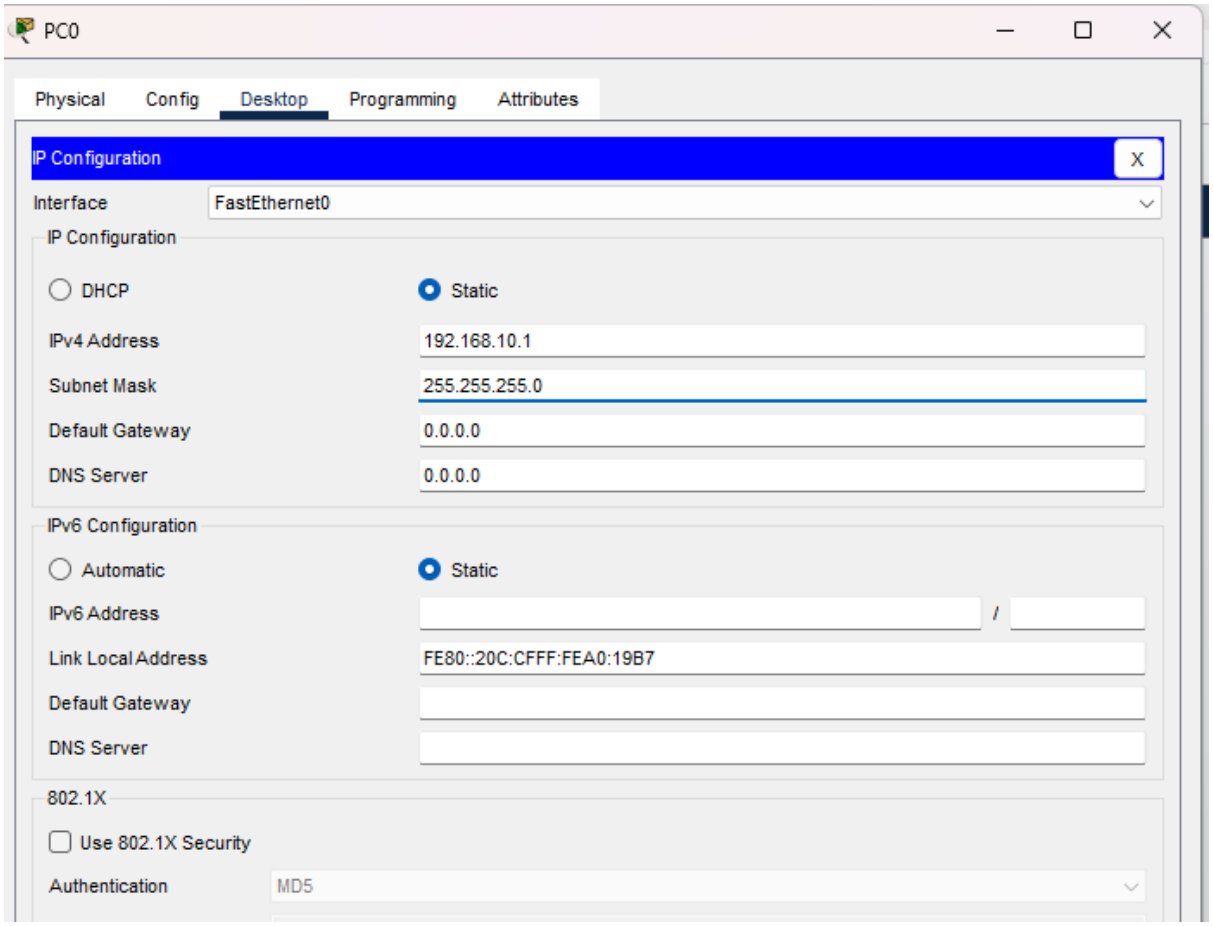

2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1



PART 2: Configure IP Addresses

Step 4: Assign IP to PC0

1. **Double-click on PC0**
2. Click the **"Desktop"** tab
3. Click **"IP Configuration"**
4. Enter the following:
    - IP Address: `192.168.1.1`
    - Subnet Mask: `255.255.255.0`
    - Default Gateway: *(leave blank)*
5. Close the window



PART 3: Add Explanation Notes to Workspace

SLIDING WINDOW PROTOCOL:

- Used for Flow Control & Error Control
- Sender can send multiple frames before waiting for acknowledgment
- Window Size controls number of unacknowledged frames allowed at a time
- Improves network efficiency & throughput

Step 7: Add Note about Piggybacking

PIGGYBACKING:

- Acknowledgments are attached/included inside outgoing DATA frames
- Avoids sending separate ACK packets
- Saves bandwidth and improves efficiency
- PC1 sends ACK inside its own data frames

back to PC0

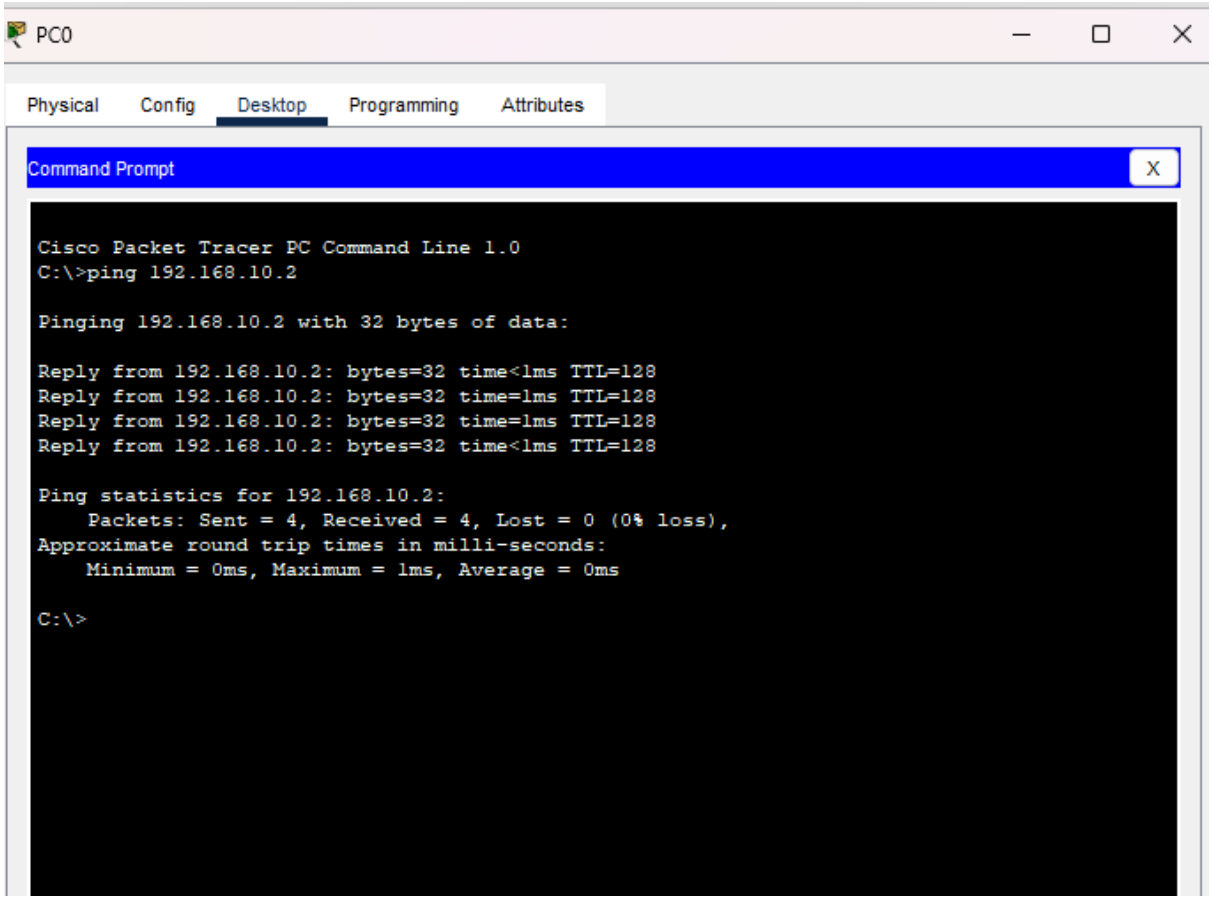
PART 4: Test Connectivity First (Realtime Mode)

Step 8: Ping from PC0 to PC1

1. **Double-click PC0**
2. Go to **Desktop** tab → click **"Command Prompt"**
3. Type the ping command:

```
ping 192.168.1.2
```

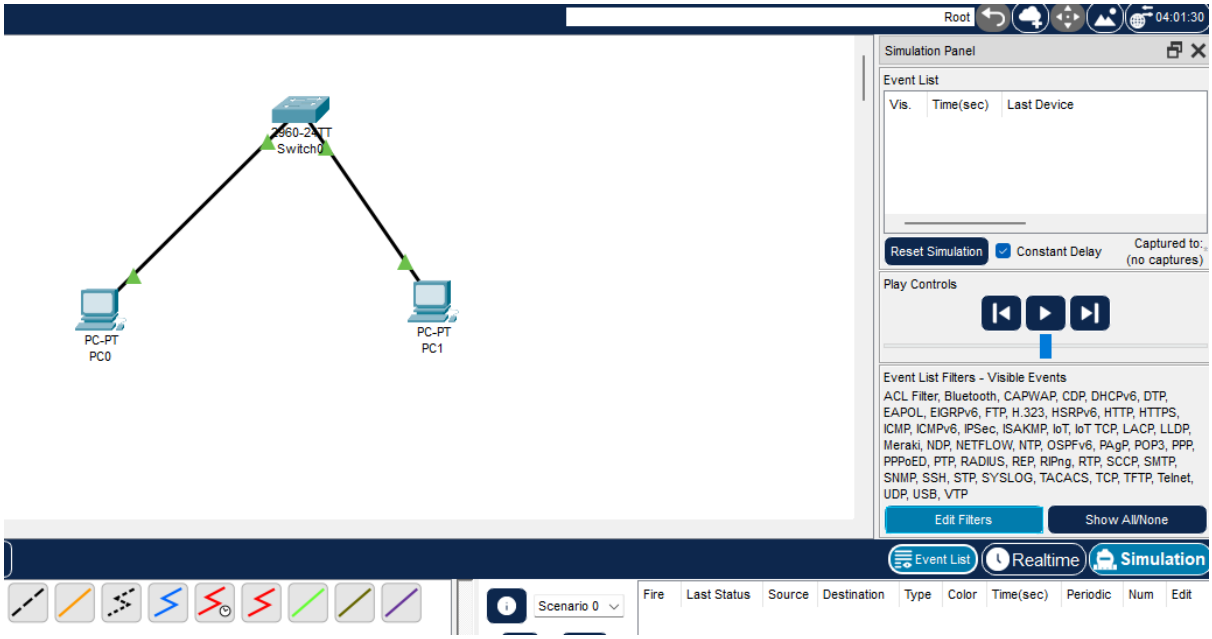
4. You should see **4 replies** from 192.168.1.1 ✅
5. This confirms the network is working correctly
6. Close the window



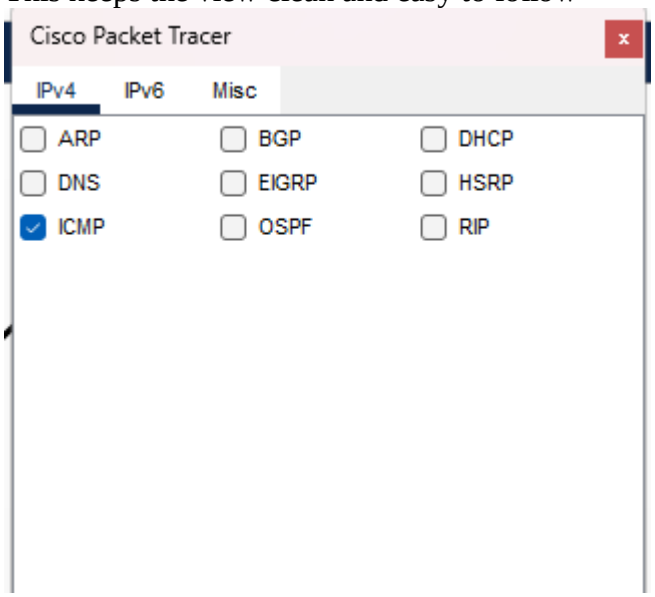
PART 5: Simulate Sliding Window with Piggybacking

Step 9: Switch to Simulation Mode

1. At the **bottom-right corner**, click **"Simulation"** button *(OR press the hourglass icon — it switches from Realtime to Simulation)*
2. The **Simulation Panel** will appear on the right side
3. You will now see packet events being captured

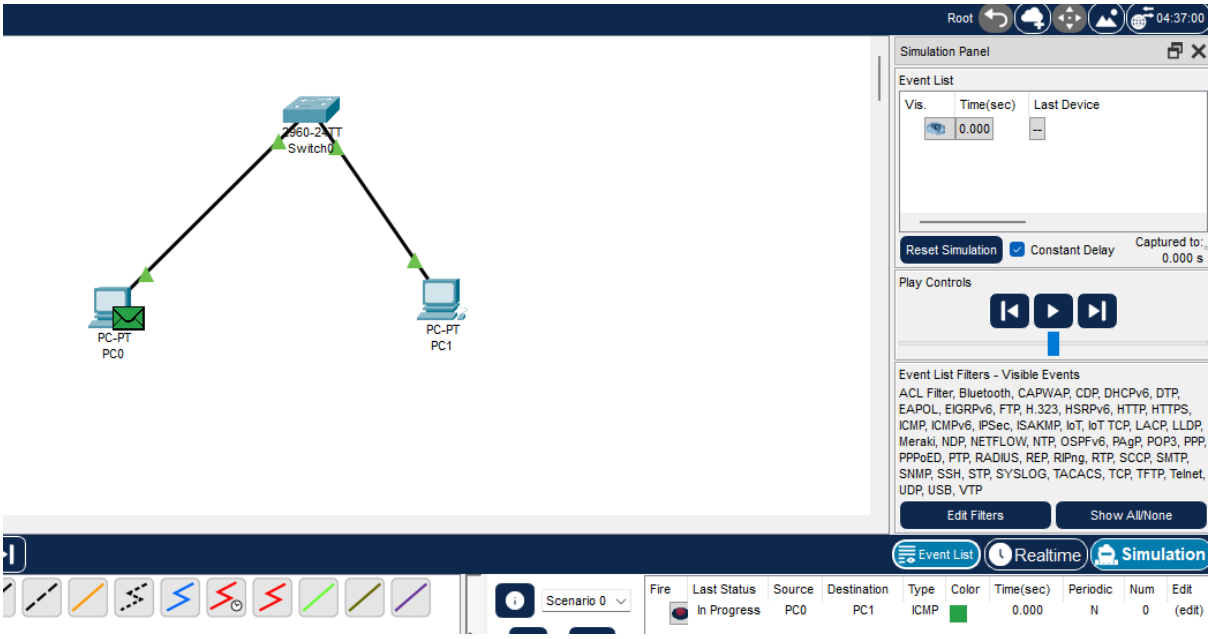


Step 10: Filter Events (Optional but Recommended)

1. In the Simulation Panel, click **"Edit Filters"**
2. Uncheck everything **except**:
   - ☑ ICMP
   - ☑ TCP (if available)
3. This keeps the view clean and easy to follow
4. 

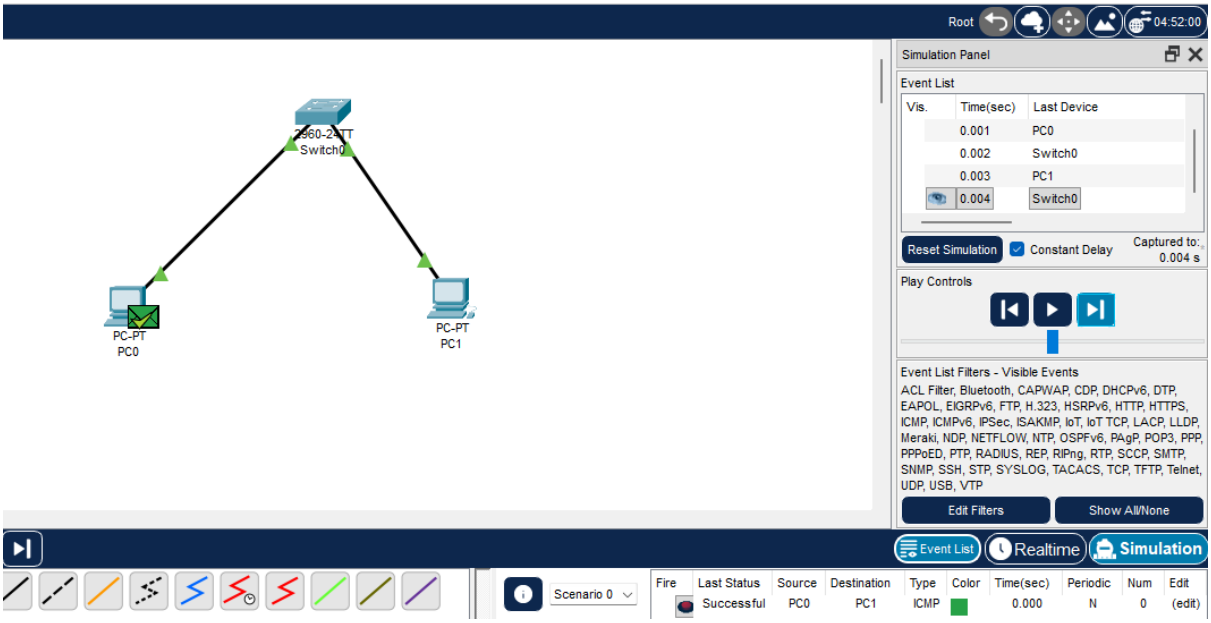
Step 11: Add Simple PDU (Data Frame PC0 → PC1)

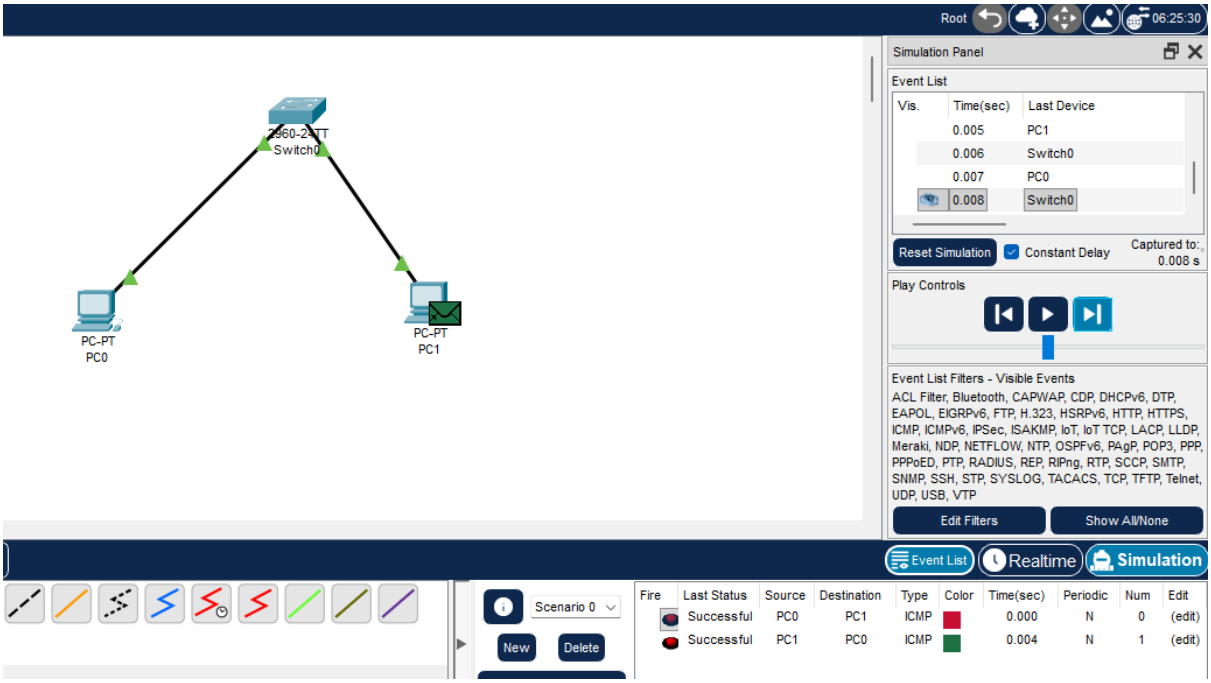
1. Click the **"Add Simple PDU"** tool *(envelope icon at the top-right toolbar)*
2. **Click on PC0** first (source)
3. **Click on PC1** second (destination)
4. A PDU (packet) will appear in the workspace
5. You'll see it listed in the **Event List** on the right

6. 

Step 12: Add Return PDU (Piggybacked ACK — PC1 → PC0)

1. Click **"Add Simple PDU"** again
2. **Click on PC1** first (source)
3. **Click on PC0** second (destination)
4. This simulates **piggybacking** — PC1 sends data back with ACK included





Step 15: Observe Piggybacked Acknowledgments

1. Watch the **return packet from PC1 to PC0**
2. This packet **carries both data AND acknowledgment** together
3. This is the **piggybacking in action** — no separate ACK packet needed